Question 1: (20 Marks)

Develop an application for managing event registrations for a fictional tech conference. The application should allow users to add new participants to the event and save the participant details to a file. Each participant entry should be saved in a structured format.

Requirements:

1. Participant Class:
   - Create a `Participant` class with the following properties:
     - `ParticipantID` (int)
     - `Name` (string)
     - `Email` (string)
2. EventManager Class:
   - Create an `EventManager` class with the following functionalities:
     - Add a new participant.

Save all participant details to a file named `registrations.txt` in the following format:

```
ParticipantID: 1, Name: Jane Doe, Email: jane.doe@example.com
ParticipantID: 2, Name: John Smith, Email: john.smith@example.com
```

3. Program Flow:
   - The application should:
     - Prompt the user to enter details for a new participant (ParticipantID, Name, Email).
     - Add the new participant to a collection of participants.
     - Save the participant details to the file `registrations.txt` each time a new participant is added.

Tasks:

1. Implement the `Participant` class. (5)
2. Implement the `EventManager` class with methods to add a new participant and save participant details to the file. (10)
3. Implement the main program flow to interact with the user and utilize the `EventManager` class.(5)

```
using System;
using System.Collections.Generic;
using System.IO;

namespace EventRegistration

class Participant
{
```

public int ParticipantID { get; set; }

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```
public string Name { get; set; }
public string Email { get; set; }

}

class EventManager
{
    private List<Participant> participants = new List<Participant>();   → if need of separate storage from files

    public void AddParticipant(Participant participant)
    {
        StreamWriter writer = new StreamWriter("registrations.txt", append: true);
        string JSONParticipant = JsonSerializer.Serialize(participant);
        writer.WriteLine(JSONParticipant);
        writer.Close();
    }

        Participants.Add(participant);
        SaveParticipantToFile(participant);

    private void SaveParticipantToFile(Participant participant)
    {
        StreamWriter writer = new StreamWriter("registrations.txt", append: true);
        string JSONParticipant = JsonSerializer.Serialize(participant);
        writer.WriteLine(JSONParticipant);
        writer.Close();
    }
}

class Program
{
    static void Main(string[] args)
    {
        EventManager eventManager = new EventManager();

        while (true)
        {
            int ID;
            Console.WriteLine("Enter the Participant Id: ")
ID = int.Parse(Console.ReadLine());

            string name;
            Console.WriteLine("Enter the Participant name: ");
```

```
            name = Console.ReadLine();
            Console.WriteLine("Enter Email: ");
            string email;
            email = Console.ReadLine()
            Participant P = new Participant { ParticipantID = ID, Name = name, Email = email};
            eventManager.AddParticipant(P);


            Console.WriteLine("Participant added and saved to file.");
        }
    }
  }
}
```

26

Question 2 (15 marks)

You are developing an ASP.NET Core MVC application for managing a shopping cart. The application allows users to add items to their cart and view the contents of the cart. Fill in the missing code in the following controller.

```
using identiryFinalExamQuestion.Models;
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;
using System;
using System.Collections.Generic;
using System.Text.Json;

public class CartController : Controller
{
  [HttpGet]
  public IActionResult Index()
  {
    var cartItems = GetCartFromSession();
    return View(cartItems);
  }

  [HttpGet]
  public IActionResult AddToCart()
  {
    var cartItems = new List<CartItem>
    {
      new CartItem { Id = 1, Name = "Laptop", Price = 999.99m },
      new CartItem { Id = 2, Name = "Phone", Price = 499.99m }
    };

    return View(cartItems);
```


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```
}

[HttpPost] 1
public IActionResult AddToCart(List<CartItem> items)
{
    var cartItems = GetCartFromSession();
Add code here to add the items in to the cart (3)
        foreach (CartItem item in items)
        {
            cartItems.Add(item);
        }
        // new cartItems list has all record
        // saving it to session

    SaveCartToSession(cartItems);
    return RedirectToAction("Index");
}


[HttpGet]
public IActionResult ClearCart()
{
Add code to clear the cart (1)
    return RedirectToAction("Index");
}


private List<CartItem> GetCartFromSession()
{
Add code to get the data from the cart (5)
        if (HttpContent.Session.Key.Contains("cartItems"))
        {
            string cart = HttpContent.Session.GetString("cartItems");
            items = JsonSerializer.Deserialize<List<CartItem>>(cart);

            return items;
        }
        else
        {
            return items;
        }
}

private void SaveCartToSession(List<CartItem> cartItems)
{
Save cart data into session (5)
        if (HttpContent.Session.Key.Contains("cartItems")) OR (!HttpContent.Session.KeyContains("cartItems"))
        {
            string jsonCart = JsonSerializer.Serialize(cartItems);
            HttpContext.Session.SetString("cartItems", jsonCart);

        }
```

→ To read prev cart (code is in SaveCartToSession)

```
        else
        {
            string alreadySavedCart = HttpContext.Session.GetString("cartItems");
            string JsonCart = JsonSerializer.Serialize(CartItems);
            string completeCart = alreadySavedCart + JsonCart;
            HttpContext.Session.SetString("cartItems", completeCart);
        }
    }
}
```

Question # 3 (10 marks)

Consider the following ASP.NET Core MVC Controller:

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Http;
using System;

namespace CookieExample.Controllers
{
    public class CookieController : Controller
    {
        public IActionResult Index()
        {
            // Set a cookie
            CookieOptions options = new CookieOptions
            {
                Expires = DateTime.Now.AddMinutes(5),
                HttpOnly = true
            };
```


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```
Response.Cookies.Append("SessionID", "12345", options);
Response.Cookies.Append("UserName", "JohnDoe", options);

// Get the cookie values
string sessionId = Request.Cookies["SessionID"];
string userName = Request.Cookies["UserName"];

// Modify the UserName cookie
if (userName != null)
{
    userName = userName.ToUpper();
    Response.Cookies.Append("UserName", userName, options);
}

// Add a new cookie if it doesn't exist
if (Request.Cookies["VisitCount"] == null)
{
    Response.Cookies.Append("VisitCount", "1", options);
}
else
{
    int visitCount = int.Parse(Request.Cookies["VisitCount"]);
    visitCount++;
    Response.Cookies.Append("VisitCount", visitCount.ToString(), options);
}
```


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```
        // Remove the SessionID cookie
        Response.Cookies.Delete("SessionID");

        // Get the modified cookie values
        string modifiedUserName = Request.Cookies["UserName"];
        string visitCountValue = Request.Cookies["VisitCount"];
        string sessionIdAfterDeletion = Request.Cookies["SessionID"];

        // Return the cookie values to the view
        ViewBag.SessionID = sessionId;
        ViewBag.UserName = userName;
        ViewBag.ModifiedUserName = modifiedUserName;
        ViewBag.VisitCount = visitCountValue;
        ViewBag.SessionIDAfterDeletion = sessionIdAfterDeletion;

        return View();
    }
  }
}
```

What will be the value of `ViewBag.SessionID` when the `Index` action is executed?

12345

What will be the value of `ViewBag.UserName` when the `Index` action is executed?

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What will be the value of `ViewBag.ModifiedUserName` when the `Index` action is executed?

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What will be the value of `ViewBag.VisitCount` when the `Index` action is executed on the first visit? What will it be on the second visit?

1 on first visit
2 on second visit

What will be the value of `ViewBag.SessionIDAfterDeletion` when the `Index` action is executed?

null

Question # 4 (10 marks)

Consider the following model, controller, and view and answer the given question.

Models/Order.cs

```
public class Order
{
    public int Id { get; set; }
    public string ProductName { get; set; }
    public int Quantity { get; set; }
    public decimal Price { get; set; }
    public decimal TotalAmount => Quantity * Price;
}
```

Controllers/OrderController.cs

```
public class OrderController : Controller
{
    [HttpGet]
    public IActionResult Create()
    {
        return View();
    }

    [HttpPost]
    public IActionResult Create(Order order)
    {
        // Simulate saving the order to the database
        TempData["SuccessMessage"] = $"Order for {order.Quantity} {order.ProductName}(s) placed
successfully!";
        return RedirectToAction("Details", new { id = 1 });
    }

    [HttpGet]
    public IActionResult Details(int id)
```


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```
    {
        // Simulate retrieving the order from the database
        var order = new Order
        {
            Id = id,
            ProductName = "Laptop",
            Quantity = 2,
            Price = 999.99m
        };

        return View(order);
    }
}
```

Views/Order/Create.cshtml:

```
@model Order
```
asp-controller="Order"

```
<form asp-action="Create" method="post">

    <div>

        <label asp-for="ProductName"></label>

        <input asp-for="ProductName" />

    </div>

    <div>

        <label asp-for="Quantity"></label>

        <input asp-for="Quantity" />

    </div>

    <div>

        <label asp-for="Price"></label>

        <input asp-for="Price" />

    </div>

    <button type="submit">Create</button>

</form>
```


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Views/Order/Details.cshtml:

```
@model Order
<h2>Order Details</h2>
<p>Product Name: @Model.ProductName</p>
<p>Quantity: @Model.Quantity</p>
<p>Price: @Model.Price</p>
<p>Total Amount: @Model.TotalAmount</p>
@if (TempData["SuccessMessage"] != null)
{
    <p>@TempData["SuccessMessage"]</p>
}
```

A user navigates to `/Order/Create` and fills out the form with the following values:

- ProductName: "Tablet"
- Quantity: 3
- Price: 300.00

The user submits the form.

What will the user see if the form is successfully submitted?

Order Details
Product Name: Laptop
Quantity : 2
Price : 999.99
Total Amount : 1999.98
Order for 3 Tablets placed successfully!

10

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